

CURRICULUM VITAE – THOM MOTT

PERSONAL INFORMATION

Thom Mott
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SKILLS

Operating systems: Linux

Programming languages: C, C++, C#, HLSL, JAVA, JavaScript, TypeScript, Python, Nix, RISC-V Assembly, Rust, Zig

Engines: Unity, Godot, Phaser, Dear ImGui

Languages: English

Relevant Courses: Parallel Programming (CSE 113), Distributed Systems (CSE 138), Introduction to Computer Graphics (CSE160), Advanced Programming (CMPM 123), Creative Coding (CMPM 169), Game AI (CMPM 146), Principles of Computer Systems Design (CSE 130), User Interface and User Experience Design (CMPM 130)

EDUCATION

University of California, Santa Cruz Santa Cruz, CA *September 2026-Future*
Degree in Progress: Computational Media (M.S.)

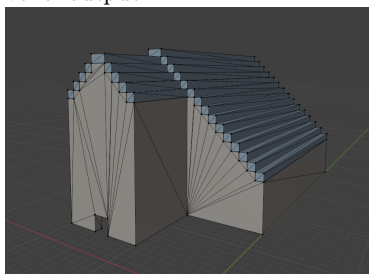
University of California, Santa Cruz Santa Cruz, CA *June 2024-June 2026*
Degree: Computer Science: Game Design (B.S.)
Dean's List: Fall 2024, Winter 2025, Spring 2025, Fall 2026

Cabrillo College Aptos, CA *August 2022-August 2024*

EMPLOYMENT HISTORY

Research Assistant, SLOTHLab UC, Santa Cruz *September 2025-Present*

- Worked alongside other undergrad and graduate researchers on a submission to the annual Generative Design in Minecraft (GDMC) competition.
- Extended Dr. Eger's procedural generation system ([Box-Split Grammars, Eger 2022](#)) by refactoring the codebase to fully typed modern Python, decoupling grammar evaluation from output via a declarative scope subclass, and adding a trimesh 3D mesh backend alongside the original Minecraft voxel output.



Undergrad Course Assistant UC, Santa Cruz *January 2025-Present*

- **Relevant Courses:** Data Structures & Algorithms (C++, Python), Game Development Experience (JS), Game Development Patterns (C#, TS), Game Technologies (C#), Writing for Game Technologies, Game Systems
- Ran in-person and online office hours to aid students in upper-division Computer Science/Computational Media courses.
- Helped to transition a C/C++ Data Structures and Algorithms course into modern idiomatic Python (static typing, generics, comprehensions). This included 4 programming assignments, 12 assessments (2,000 lines)
- Modernised course tech stacks by transitioning legacy Docker & Linux environments to modern tooling.
- Assisted professors with course material preparation.
- Hosted & coordinated section, assisted in proctoring exams, graded student papers and exams.

Research Assistant, SET Lab UC, Santa Cruz*September 2024-July 2025*

- Contributed to a research-driven AR game focused on climate resilience education and community engagement, working alongside a postdoctoral researcher and undergraduate research assistants.
- Developed UI components and a location-based overworld map in Unity, and provided technical guidance to team members on engine usage and development practices.

Intern, Guardian Gamer Los Gatos, CA*June 2024-September 2024*

- Built an internal admin panel for user and group management, supporting flexible lookup by multiple account identifiers and real-time modification of records, with data sourced and processed from AWS-managed databases.

PROJECTS

Gamba Gun Programmer, Designer*June 2025-Present*

- Developed a GPU-based splat rendering system enabling high-density, dynamic surface effects (e.g. fire propagation), improving visual fidelity and gameplay expressiveness.
- Implemented a localisation system to support scalable multilingual content.
- Contributed to refactoring and long-term maintainability of core gameplay systems in an active commercial project.

California V2 Producer, Designer, Artist*March 2020-Present*

- Led production for a 6-person international team over six years on a large-scale curated map for “Unturned”.
- Designed major locations, NPC-driven quest content, and gameplay balance across survival systems.
- Created hundreds of optimised 3D assets to support a cohesive open-world environment.
- Organised and coordinated large-scale playtests across multiple countries and time zones.

Computer Graphics Showcase (CSE160) Creator*Winter 2025*

- Learned basics of OpenGL & computer graphics technologies by implementing several systems, non-deformable mesh rigging and key-frame animation and blinn-phong lighting models.

Wang Slime Creator*November 2025*

- Implemented a fully GPU-driven slime mould simulation operating over a procedurally generated terrain with compute shaders.
- Integrated Wang tile-based world generation with live runtime updates, allowing dynamic modification of the simulation domain.
- Focused on performance-oriented design and parallel execution.

NixOS Configuration Creator*March 2025*

- Used Nix to declaratively manage Linux environment across multiple PCs.
- Managed virtual machines using Nix.
- Packaged software non-nix software for usage in NixOS.

Sistor Fixor Creator*November 2025*

- Developed a digital logic puzzle game prototype in Unity within a two-day timeframe for CMPM170.
- Implemented a simplified but internally consistent logic simulation system focused on problem-solving rather than real-world electronics accuracy.
- Designed timed circuit-based challenges with clear input/output constraints.

Dungeon Generator Creator*June 2025*

- Built a performant procedural dungeon generator using a backtracking-style algorithm with weighted random selection for CMPM146.
- Designed modular room connectivity inspired by “The Binding of Isaac”.
- Emphasised deterministic generation and runtime efficiency.

Spebby's Chess Engine Programmer*November-December 2024*

- Implemented a chess engine and AI from scratch in C++, using bitboards for efficient board representation.
- Developed a Negamax-based AI with heuristic evaluation, capable of real-time play to depth 5.
- Built a graphical frontend using Dear ImGui.

Booliards Programmer*April 2023*

- Contributed to a fast-paced arcade billiards game developed in Unity.
- Worked on game and wave design, achievements, and backend architecture.
- Integrated platform services including Newgrounds APIs.

Egg Rush Programmer*July 2021*

- Developed core gameplay systems, enemy behaviours, and upgrade mechanics for a beat'em-up game jam entry.
- Delivered a complete, polished experience within a three-day development window.

Tower of Towers Programmer

March-May 2021

- Implemented core gameplay features and level content for a 16-bit-style platformer.
- Wrote localisation support and platform integration.

Easter Island Lead, Designer, Artist

February 2020

- Led development an officially endorsed curated map for Unturned, featured in-game.
- Focused on level design, gameplay balancing, project coordination, and 3D asset creation.
- Supported through post-release content updates and maintenance, including bug fixes and optimisation.